Impact of Oil Price Shocks on the U.S. Stock Market

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Summary

This paper replicates and extends Kilian and Park (2009) to examine the relationship between oil market shocks and U.S. real stock returns. The analysis distinguishes between oil supply, aggregate demand, and oil-specific demand shocks and assesses whether the original findings remain robust to changes in data, model specification, stock return measures, and sample periods. The narrow replication retains the original structural VAR specification while using alternative data sources, revised measures of global real activity, and a different measure of U.S. stock returns. The broader replication extends the analysis using local projections, compares price and total return measures of the S&P 500, and lengthens the sample to include post-2006 data. The narrow replication produces results broadly consistent with the original study, although several responses differ in their timing and magnitude. The broader analysis finds that estimated stock market responses are sensitive to model choice, the treatment of dividends, and the sample period, while oil price responses are comparatively robust. Overall, the central finding of Kilian and Park (2009) remains supported: oil demand shocks account for a substantial share of the variation in U.S. real stock returns.

1. Introduction

Oil price shocks are recognised as an important driver of financial markets. Kilian and Park (2009) showed that structural oil demand and precautionary demand shocks, rather than supply shocks, account for much of the variation in U.S. real stock returns. Using a recursively identified (Cholesky) structural VAR, they also demonstrated that the effects of these shocks depend on its type.

This paper replicates their analysis in both a narrow and broad sense. The narrow replication uses the same model specification but relies on alternative data sources, revised global real activity measures, and different stock return series. The broad replication extends the analysis by employing a Local Projection model, distinguishing between price and total return measures of stock returns, and lengthening the sample to include post-2006 data. These extensions allow an assessment of how far the original results are robust to differences in data and model choice.

The narrow replication produces results largely consistent with Kilian and Park (2009), though some impulse responses differ in timing and magnitude. The broader replication, however, shows more variation. In particular, the response of U.S. real stock returns to aggregate demand shocks depends on whether the structural VAR or Local Projection is used, on whether dividends are included in the return measure, and on the sample period considered. By contrast, the responses of oil prices to the three structural shocks are relatively robust. Overall, the key result in Kilian and Park (2009) – that oil demand shocks account for a substantial share of the variability in U.S. real stock returns – continues to hold.

2. Data

The data includes a measure of the percent change in world crude oil production, the real price of oil imported by the U.S., the aggregate U.S. real stock return, and an indicator of global real activity. All data used are monthly. Two main sample periods are considered: 1973.12-2006.12, which follows the original study (excluding the first 11 months due to data availability), and 2006.12-2025.3 as an extension.

The percent change in world crude oil production is constructed by taking the first difference of logs of crude oil including lease condensate (from the U.S. Energy Information Administration) and scaling by 100. The real price of oil is constructed by deflating the U.S. imported refiner's acquisition cost of crude oil (from the U.S. Energy Information Administration) by the U.S. Consumer Price Index (from the U.S. Bureau of Labour Statistics) and scaling by 100.

Two measures of the aggregate U.S. real stock return are used: the log return of the S&500 price index minus the CPI inflation rate, and the log return of the S&500 total return index minus the CPI inflation rate, each scaled by 100. The Index of Global Real Activity from the Federal Reserve Bank of Dallas is used as an indicator of global real activity. The data for both S&500 indices are sourced from Investing.com.

3. Model Specification

Three models are considered. The first follows the structural VAR(24) model used in Kilian and Park (2009) given by

$$\boldsymbol{A}_0\boldsymbol{z}_t = \boldsymbol{\alpha} + \sum_{i=1}^{24} \boldsymbol{A}_i\boldsymbol{z}_{t-i} + \boldsymbol{\epsilon}_t$$

where $\boldsymbol{\epsilon}_t \sim N(\boldsymbol{0}, \boldsymbol{I})$ is an $n \times 1$ vector of serially uncorrelated structural shocks. $\boldsymbol{z}_t$ is an $n \times 1$ vector of dependent variables that contains the percent change in log oil production, real activity, real oil price, and real stock returns. $\boldsymbol{\alpha}$ is an $n \times 1$ vector of intercepts, and $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_{24}$ are $n \times n$ matrices of VAR coefficients. Let the reduced form VAR residuals $\boldsymbol{e}_t \sim N(\boldsymbol{0}, \boldsymbol{\Sigma})$ be defined as $\boldsymbol{e}_t = \boldsymbol{A}_0^{-1}\boldsymbol{\epsilon}_t$ which implies $\boldsymbol{\Sigma} = \boldsymbol{A}_0^{-1}(\boldsymbol{A}_0^{-1})'$. The model imposes the following identifying assumptions[1] on $\boldsymbol{A}_0^{-1}$ resulting in a recursively identified (Cholesky) structural model of the form

$$\boldsymbol{e}_t \equiv \begin{pmatrix} e_{1t}^{\Delta global\ oil\ production} \\ e_{2t}^{global\ real\ activity} \\ e_{3t}^{real\ price\ of\ oil} \\ e_{4t}^{U.S.\ stock\ returns} \end{pmatrix} = \begin{bmatrix} \alpha_{11} & 0 & 0 & 0 \\ \alpha_{21} & \alpha_{22} & 0 & 0 \\ \alpha_{31} & \alpha_{32} & \alpha_{33} & 0 \\ \alpha_{41} & \alpha_{42} & \alpha_{43} & \alpha_{44} \end{bmatrix} \begin{pmatrix} \epsilon_{1t}^{oil\ supply\ shock} \\ \epsilon_{2t}^{aggregate\ demand\ shock} \\ \epsilon_{3t}^{oil-specific\ demand\ shock} \\ \epsilon_{4t}^{other\ shocks\ to\ stock\ returns} \end{pmatrix}$$

The second is a restricted bivariate version of the structural VAR(24) model above with the percent change in log oil production and global real activity excluded. The re-identified (Cholesky) structural model imposes the following identifying assumptions

$$\boldsymbol{e}_t \equiv \begin{pmatrix} e_{1t}^{real\ price\ of\ oil} \\ e_{2t}^{U.S.\ stock\ returns} \end{pmatrix} = \begin{bmatrix} \alpha_{11} & 0 \\ \alpha_{21} & \alpha_{22} \end{bmatrix} \begin{pmatrix} \epsilon_{1t}^{real\ oil\ price\ shock} \\ \epsilon_{2t}^{other\ shocks\ to\ stock\ returns} \end{pmatrix}$$

where the real oil price shock ϵ_{1t} is interpreted as a weighted average of the oil supply, aggregate demand, and oil-specific demand shock from the previous model.

The third model follows a Local Projection given by

$$y_{t+h} = c^{(h)} + \beta^{(h)}x_t + \boldsymbol{\gamma}^{(h)}\boldsymbol{q}_t + \boldsymbol{\phi}^{(h)}\boldsymbol{w} + v_{t+h}$$

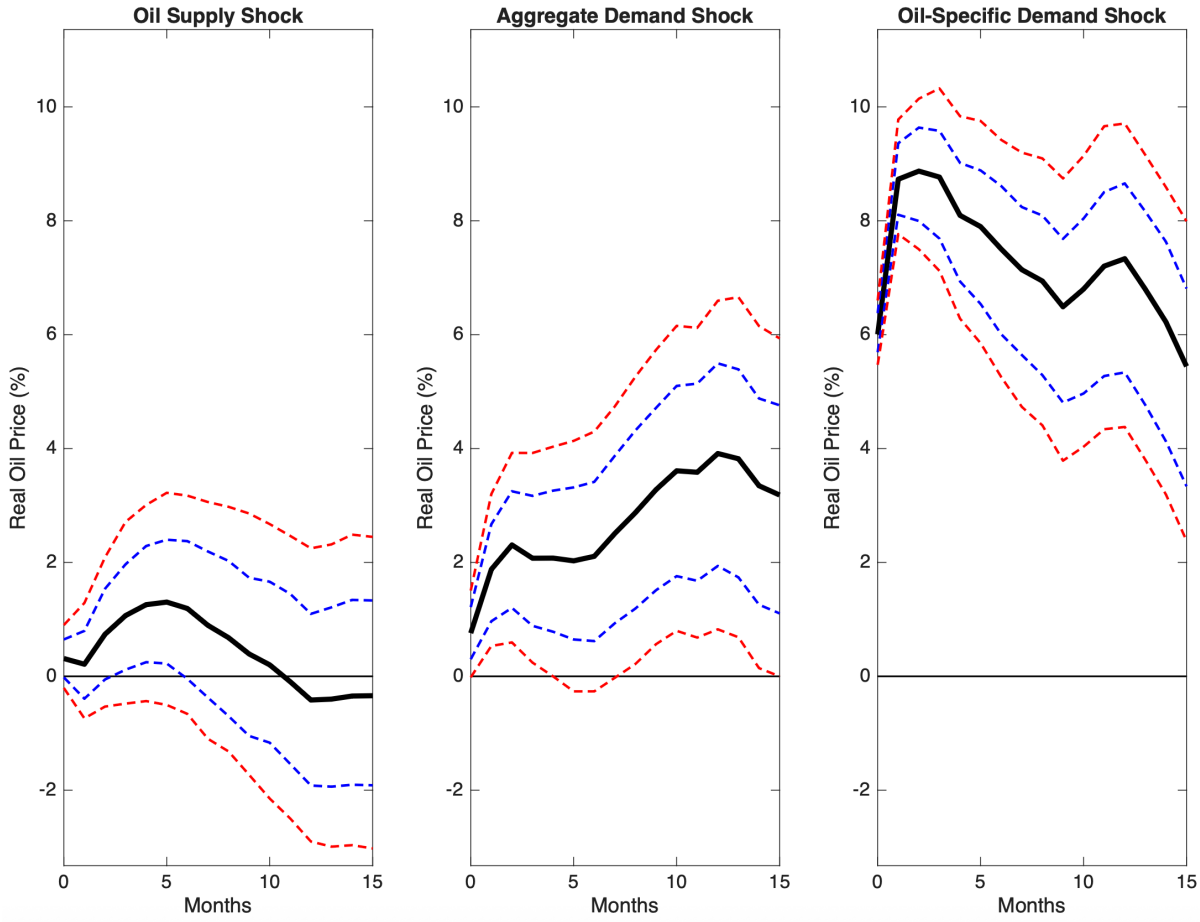
where y_{t+h} is the dependent variable of interest at time $t+h$. x_t is the shocked variable at time t, and $\beta^{(h)}$ is the impulse response of y_{t+h} to that particular shock. For differenced variables, y_{t+h} is expressed as 100 times log of its undifferenced counterpart so that $\beta^{(h)}$ has a percentage interpretation. $\boldsymbol{q}_t$ is an $n_q \times 1$ vector of contemporaneous controls, where n_q denotes the number of variables that come before x_t in the Cholesky ordering of the first model. $\boldsymbol{w}$ is a $n_w \times 1$ vector of lagged controls, containing all variables in the system lagged up to p periods so that $n_w = n \times p$. Following the structural VAR(24) model above, $\boldsymbol{w}$ will include all 24 lags of the percent change in log oil production, real activity, real oil price, and real stock returns.

[1] See Kilian and Park (2009) for the supporting arguments.

4. Results

4.1 Replication

FIGURE 1
RESPONSES OF THE REAL PRICE OF CRUDE OIL TO ONE STANDARD DEVIATION STRUCTURAL SHOCKS: POINT ESTIMATES WITH ONE AND TWO STANDARD ERROR BANDS



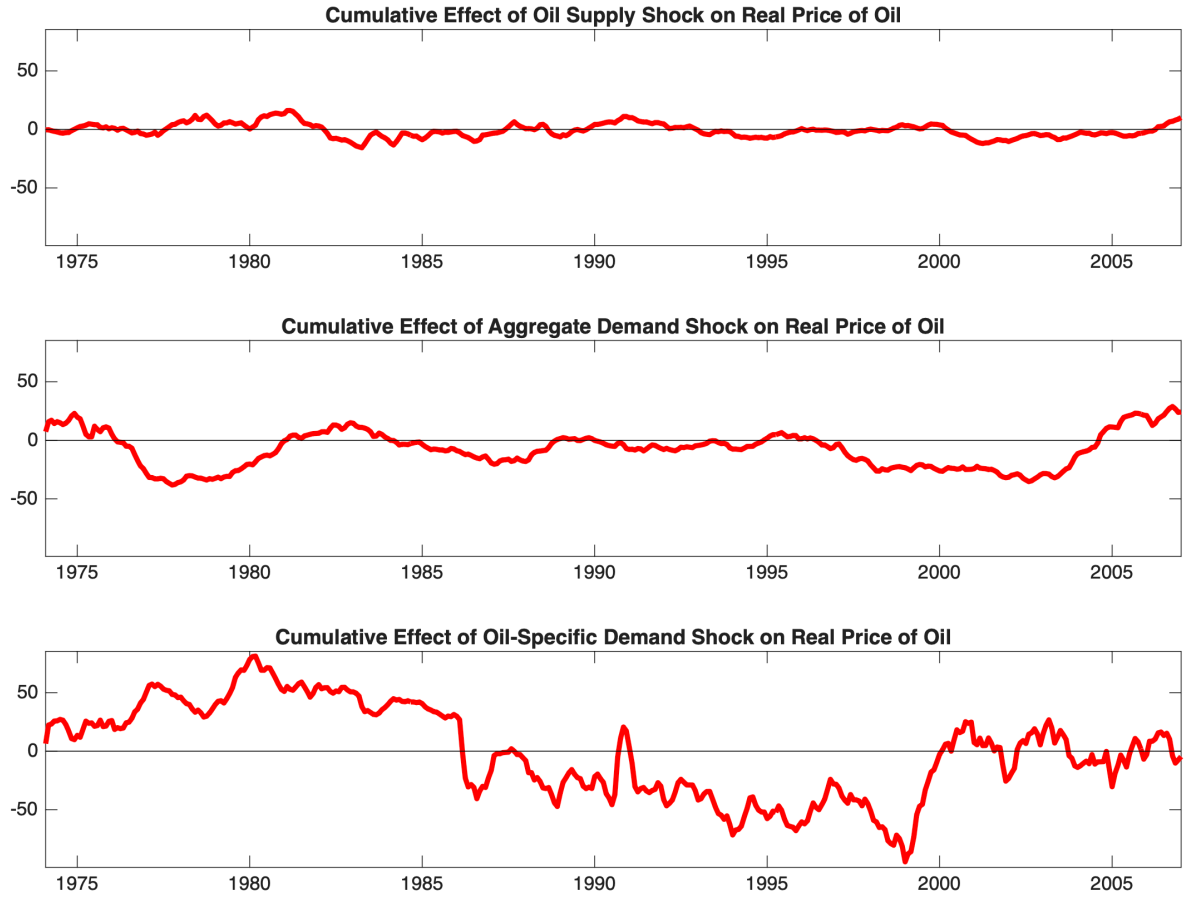
NOTES: Estimates based on the unrestricted structural VAR model. Confidence bands were constructed using bootstrap-after-bootstrap (see Kilian 1998). Oil supply shock has been normalized to represent a negative one standard deviation shock.

There are some similarities and differences between the results in Figure 1 and those in Kilian and Park (2009), potentially reflecting differences in data. First, an increase in the precautionary demand for oil causes an immediate and persistent increase in the real price of oil, which then gradually tapers off. Likewise, an unanticipated disruption in oil production causes a transitory increase in the real price of oil. Both results closely mirror those in Kilian and Park (2009). By contrast, an unexpected increase in global demand for industrial commodities produces a more immediate rise in the real price of oil, which is statistically significant for the first five months based on two standard error bands. In Kilian and Park (2009), however, this response was both weaker and more delayed.

Since the model used here is identical to that in Kilian and Park (2009), this discrepency may be attributed to (1) differences in data sources and (2) potential data revisions. The global real activity data used here were obtained from the Federal Reserve Bank of St. Louis, while Kilian

and Park (2009) used data from Drewry's Shipping Monthly, Ltd. Furthermore, this data has since been corrected for errors, as discussed in Killian and Park (2019), which may help explain the response of real oil price to an aggregate demand shock. Thus, the real oil price is more sensitive to unexpected changes in aggregate demand than was initially reported in Killian and Park (2009).

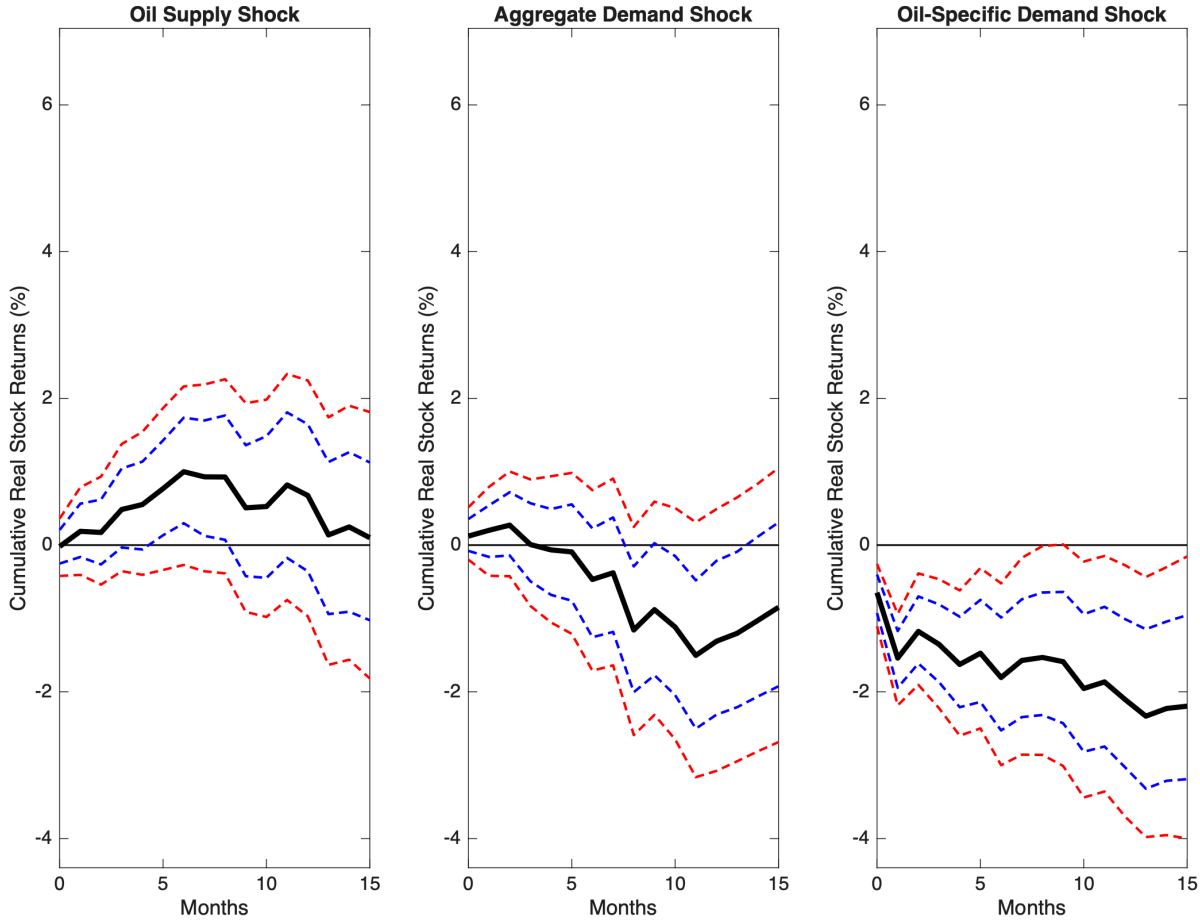
FIGURE 2
HISTORICAL DECOMPOSITION OF THE REAL PRICE OF OIL: 1974.1-2006.12



NOTES: Estimates based on the unrestricted structural VAR model.

The historical decomposition of fluctuations in the real price of oil in Figure 2 closely matches that in Kilian and Park (2009), showing that these fluctuations have been driven primarily by global and precautionary demand shocks. Specifically, precautionary demand shocks account for the majority of the variation, whereas supply shocks contribute only minimally.

FIGURE 3
CUMULATIVE RESPONSES OF SP500 REAL STOCK RETURNS TO ONE STANDARD DEVIATION STRUCTURAL SHOCKS: POINT ESTIMATES WITH ONE AND TWO STANDARD ERROR BANDS



NOTES: Estimates based on the unrestricted structural VAR model. Confidence bands were constructed using bootstrap-after-bootstrap (see Kilian 1998). Oil supply shock has been normalized to represent a negative one standard deviation shock.

There are notable differences between the results in Figure 3 and those in Kilian and Park (2009), again potentially reflecting differences in data. For example, an unanticipated disruption in oil production causes a transitory increase in U.S. real stock returns that is significant based on one standard error bands, whereas no significant effects in Kilian and Park (2009) were reported; an unexpected increase in global demand for industrial commodities causes a delayed, but sustained decrease in U.S. real stock returns, whereas the opposite direction was true in Kilian and Park (2009). The only consistent finding here is the persistently negative U.S. real stock returns in response to an increase in the precautionary demand for oil, which is statistically significant for the first 15 months.

One possible explanation is that beyond the revisions to global real activity data, as well as differences in the sources of global real activity data noted earlier, the measures of real stock returns also differ. The present analysis uses the S&P 500 index, whereas Kilian and Park (2009) used the CRSP value-weighted market portfolio. Since it was unclear whether their stock return series incorporated dividends[2], their measure may instead reflect a total return index. This could account for the positive real stock return they reported, particularly since

[2] This was only clear for their dividends series.

dividends themselves were found to rise in response to an aggregate demand shock. In contrast, the S&P 500 index used here is a price index and therefore excludes dividends from the calculation of stock returns.

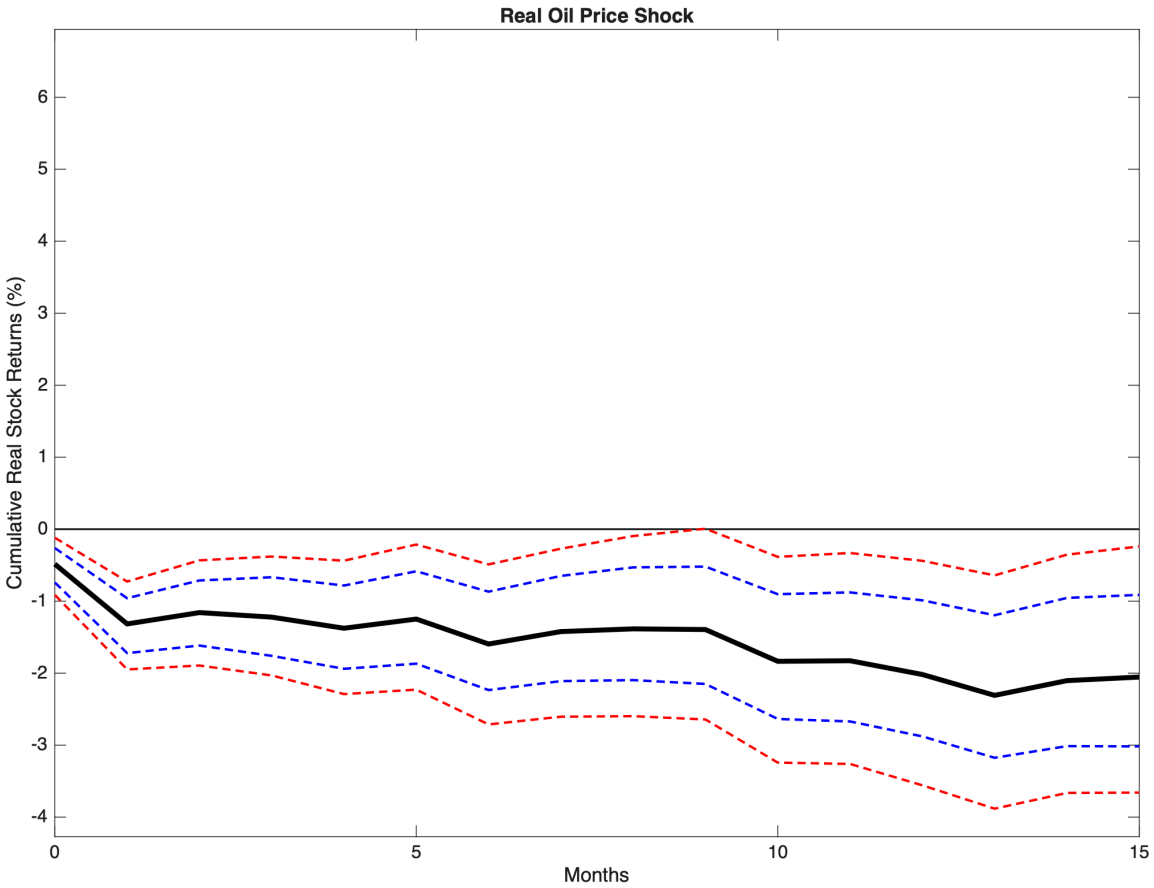
TABLE 1
PERCENT CONTRIBUTION OF DEMAND AND SUPPLY SHOCKS IN THE CRUDE OIL MARKET TO THE OVERALL VARIABILITY OF SP500 REAL STOCK RETURNS

Horizon	Oil Supply Shock	Aggregate Demand Shock	Oil-specific Demand Shock	Other Shocks
1	0.21	0.10	5.87	93.82
2	0.21	0.13	6.46	93.20
3	0.67	0.46	6.54	92.33
12	2.29	5.04	7.90	84.77
∞	6.22	7.93	9.90	75.95

NOTES: Estimates based on the variance decomposition of the unrestricted structural VAR model.

Table 1 indicates that demand shocks account for a substantially larger share of the variability in U.S. real stock returns on average than reported by Kilian and Park (2009). In the short run, only the effects of oil supply and aggregate demand shocks were negligible, whereas all three structural shocks were negligible in Kilian and Park (2009). On impact, around 6% of the variation in U.S. real stock returns is explained by oil-market specific demand shocks, suggesting that the U.S. stock market is much more sensitive and responds instantaneously to changes in the global oil market. In the long run, the three structural shocks together account for roughly 24% of the variation in U.S. real stock returns, in line with the 22% reported by Kilian and Park (2009). Among these, oil-market specific demand shocks contribute the largest share (10%), followed by aggregate demand shocks (8%) and oil supply shocks (6%). Thus, oil demand shocks collectively explain 18% of the overall variability in U.S. real stock returns, compared with only 6% for oil supply shocks. Overall, these results closely align with those of Kilian and Park (2009).

FIGURE 4
CUMULATIVE RESPONSES OF SP500 REAL STOCK RETURNS TO ONE STANDARD DEVIATION REAL OIL PRICE SHOCK: POINT ESTIMATES WITH ONE AND TWO STANDARD ERROR BANDS

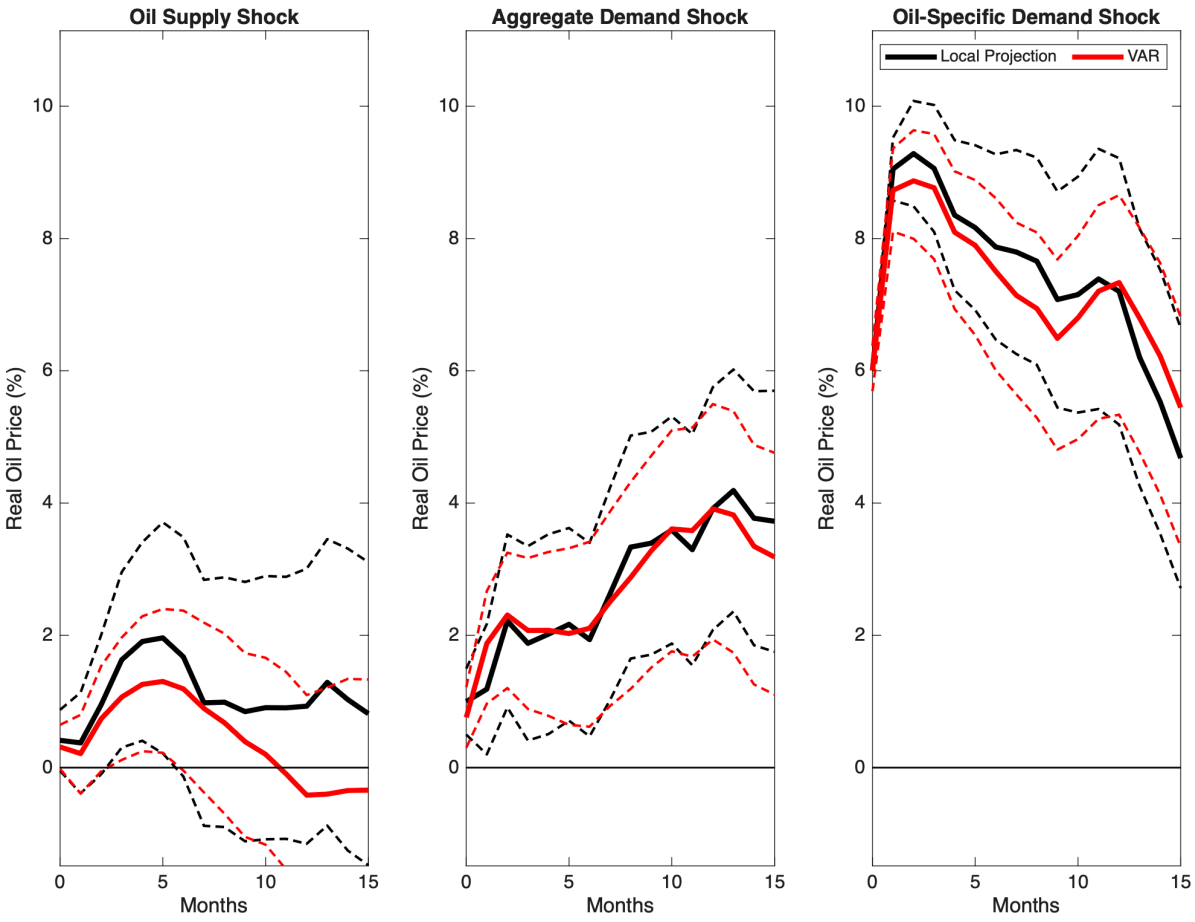


NOTES: Estimates based on the restricted bivariate structural VAR model. Confidence bands were constructed using bootstrap-after-bootstrap (see Kilian 1998).

Figure 4 shows the response of U.S. real stock returns to a real oil price shock, estimated using the restricted bivariate VAR model. The results indicate that an unanticipated increase in the real oil price leads to an unambiguous decline in stock returns. Because this shock represents a weighted combination of the three structural shocks in the unrestricted model, the response reflect their collective effect rather than the separate impact of each and should be interpreted with care. These findings are consistent with those of Kilian and Park (2009).

4.2 Local Projection

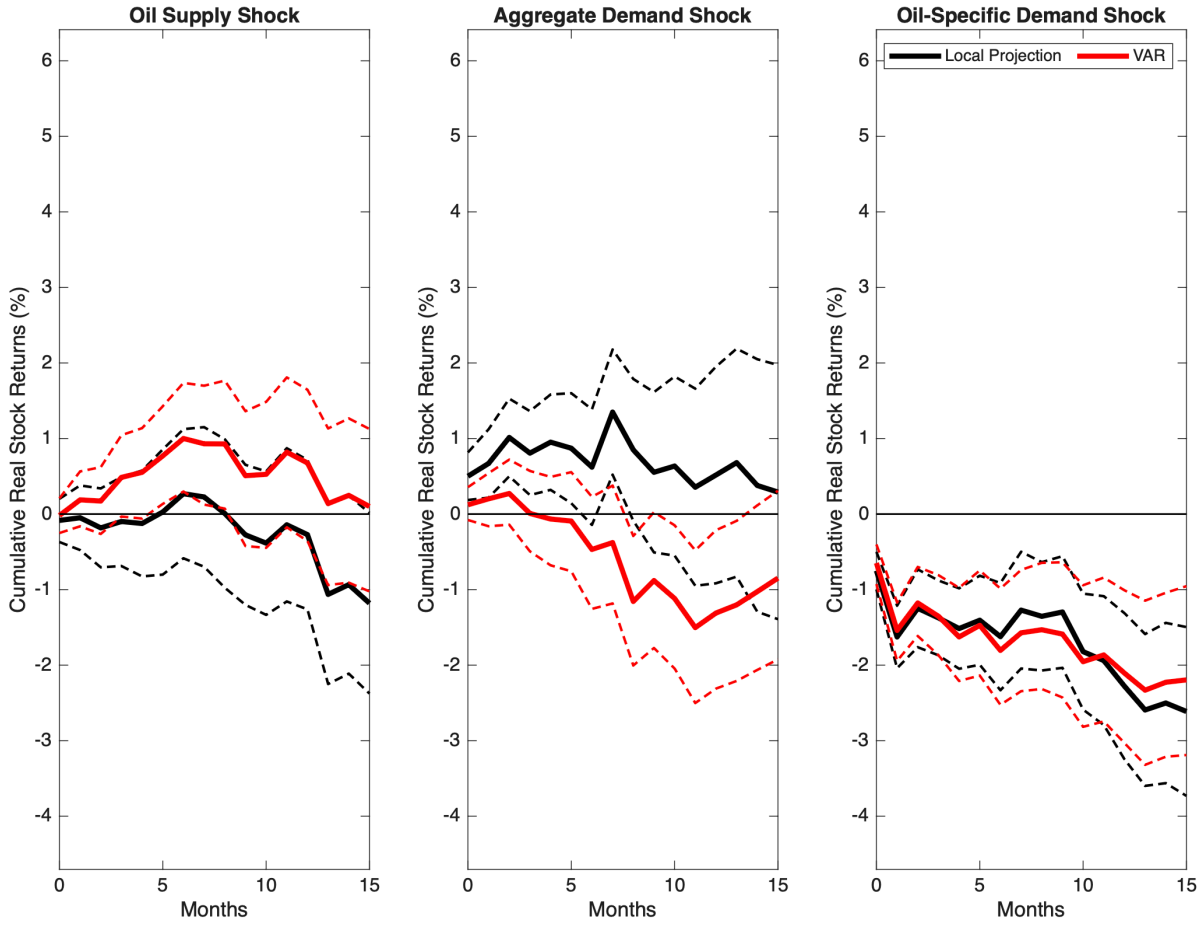
FIGURE 5
RESPONSES OF THE REAL PRICE OF CRUDE OIL TO ONE STANDARD DEVIATION STRUCTURAL SHOCKS: COMPARING STRUCTURAL VAR AND LOCAL PROJECTION POINT ESTIMATES (ONE STANDARD ERROR BANDS)



NOTES: Confidence band for the unrestricted structural VAR were constructed using bootstrap-after-bootstrap (see Kilian 1998). Confidence band for the local projections were constructed using point-wise standard errors. Oil supply shock has been normalized to represent a negative one standard deviation shock.

The Local Projection estimates of the response of the real price of oil to the three structural shocks align closely with the corresponding VAR estimates and suggest that the results discussed under Figure 1 hold under different model specifications.

FIGURE 6
CUMULATIVE RESPONSES OF SP500 REAL STOCK RETURNS TO ONE STANDARD DEVIATION STRUCTURAL SHOCKS: COMPARING STRUCTURAL VAR AND LOCAL PROJECTION POINT ESTIMATES (ONE STANDARD ERROR BANDS)

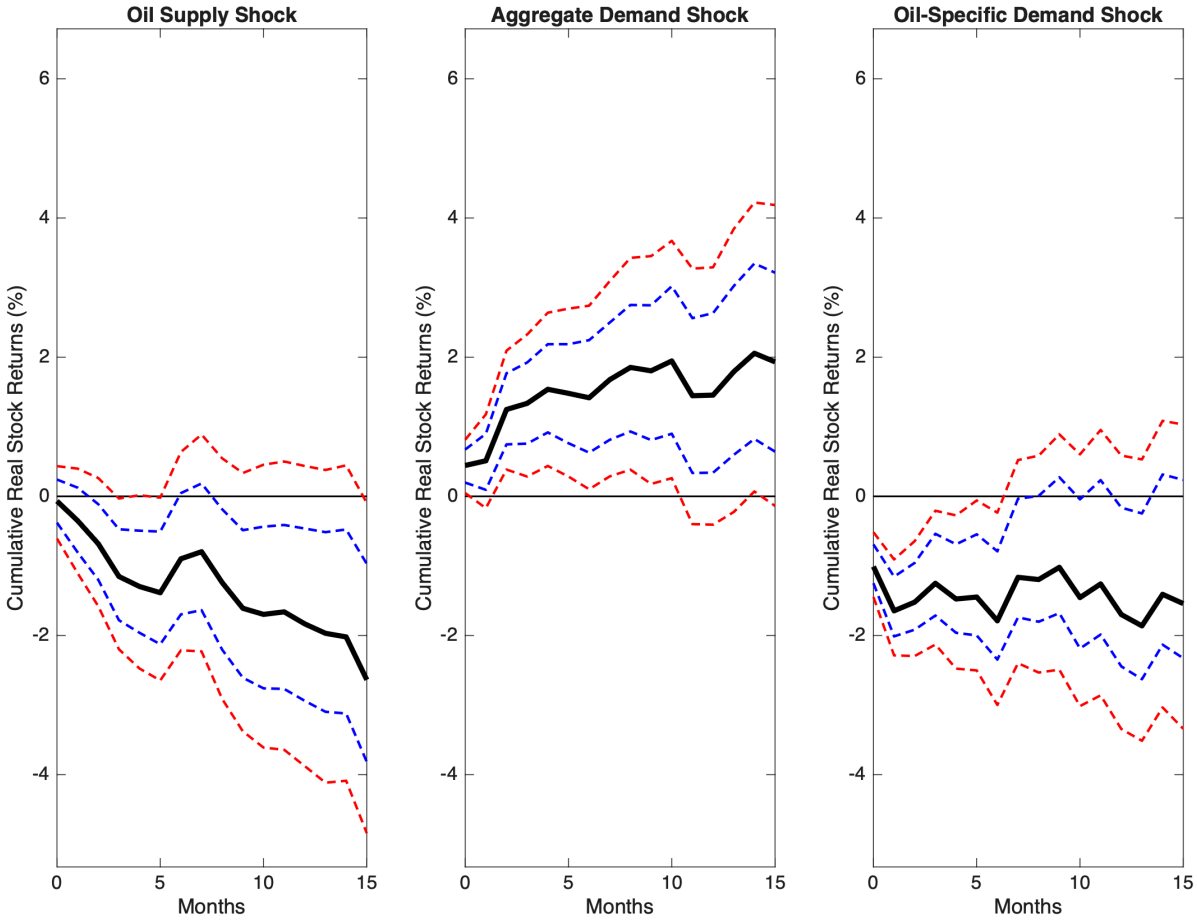


NOTES: Confidence bands for the unrestricted structural VAR were constructed using bootstrap-after-bootstrap (see Kilian 1998). Confidence bands for the local projections were constructed using point-wise standard errors. Oil supply shock has been normalized to represent a negative one standard deviation shock.

The Local Projection model successfully replicates the results of Kilian and Park (2009), which the structural VAR model could not. In particular, U.S. real stock returns now respond differently to each shock. Where the structural VAR model predicted a decline, the Local Projection model predicts a transitory increase within the first year in response to an unexpected rise in global demand for industrial commodities. Moreover, the transitory decline in U.S real stock return following an unanticipated increase in oil production is no longer statistically significant based on one standard error bands. In both models, an unexpected increase in precautionary demand consistently leads to a persistent decrease in U.S. real stock returns.

4.2 Using S&P 500 Total Return

FIGURE 7
CUMULATIVE RESPONSES OF SP500 TOTAL REAL STOCK RETURNS TO ONE STANDARD DEVIATION STRUCTURAL SHOCKS: POINT ESTIMATES WITH ONE AND TWO STANDARD ERROR BANDS

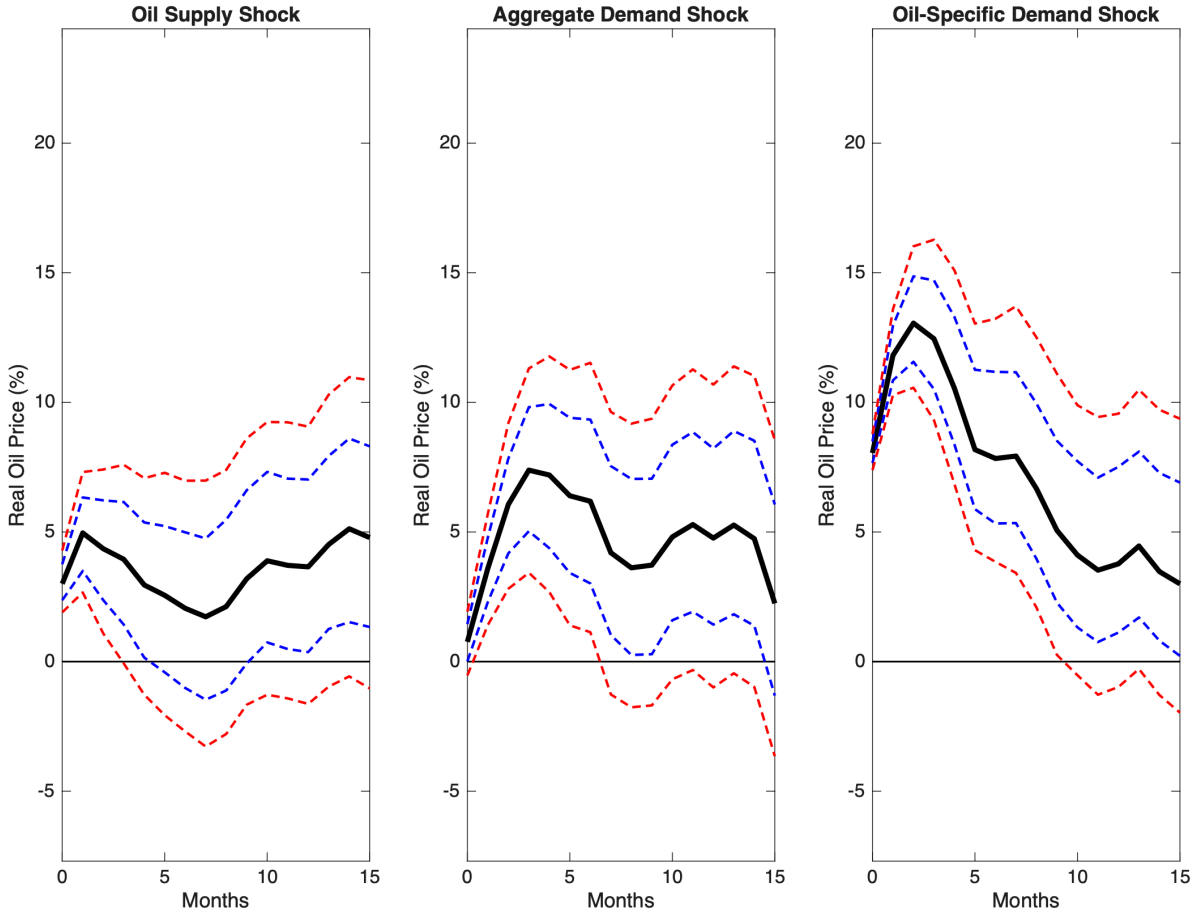


NOTES: Estimates based on the unrestricted structural VAR model using monthly data from 1988.7 to 2006.12. Confidence bands were constructed using bootstrap-after-bootstrap (see Kilian 1998). Oil supply shock has been normalized to represent a negative one standard deviation shock.

Figure 7 shows that simply replacing the S&P 500 price index with its corresponding total return index that accounts for dividends (while keeping the same structural VAR model) reproduces results similar to those of Kilian and Park (2009) and the Local Projection model above. Importantly, this strongly supports their finding that the three structural shocks have differing effects on the U.S. stock market and is not just the result of a different estimation method. Due to data availability, however, the sample period used here is limited to monthly data 1988.7-2006.12 rather than 1973.12-2006.12.

4.3 Extending the Sample Period

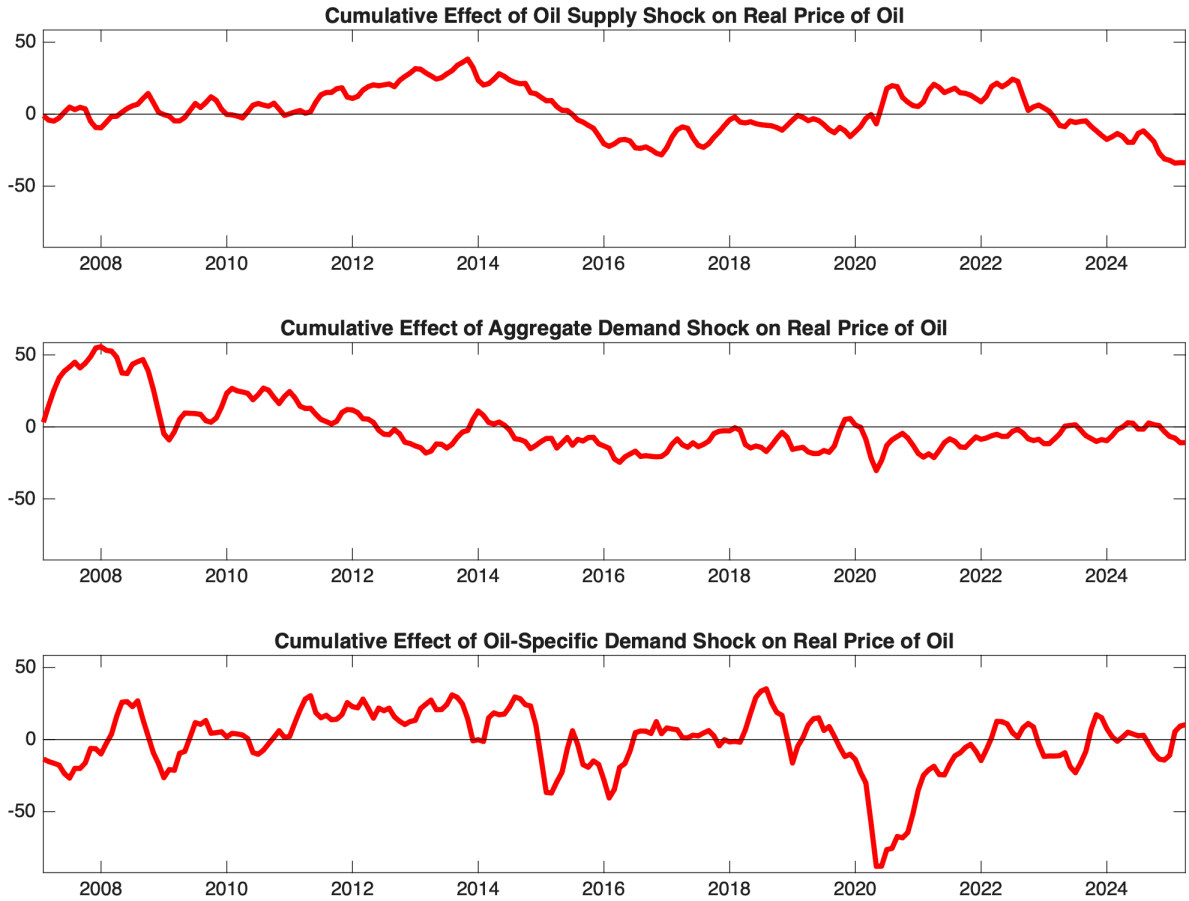
FIGURE 8
RESPONSES OF THE REAL PRICE OF CRUDE OIL TO ONE STANDARD DEVIATION STRUCTURAL SHOCKS: POINT ESTIMATES WITH ONE AND TWO STANDARD ERROR BANDS



NOTES: Estimates based on the unrestricted structural VAR model using monthly data from 2006.12 to 2025.3. Confidence bands were constructed using bootstrap-after-bootstrap (see Kilian 1998). Oil supply shock has been normalized to represent a negative one standard deviation shock.

The main finding here (Figure 8 and 9) is that the results from Figure 1 and 2 continues to hold after extending the sample period from 2006.12-2025.3. Most of the variability in real oil price is driven by global oil demand shocks, although oil supply shocks now play a larger role than before (as seen in Figure 9). Furthermore, the immediate response of the real oil price in Figure 8 to an aggregate demand shock is more pronounced than in Figure 1, indicating greater sensitivity in the current market and contrasting the delayed response reported in Kilian and Park (2009).

FIGURE 9
HISTORICAL DECOMPOSITION OF THE REAL PRICE OF OIL: 2006.12-2025.3



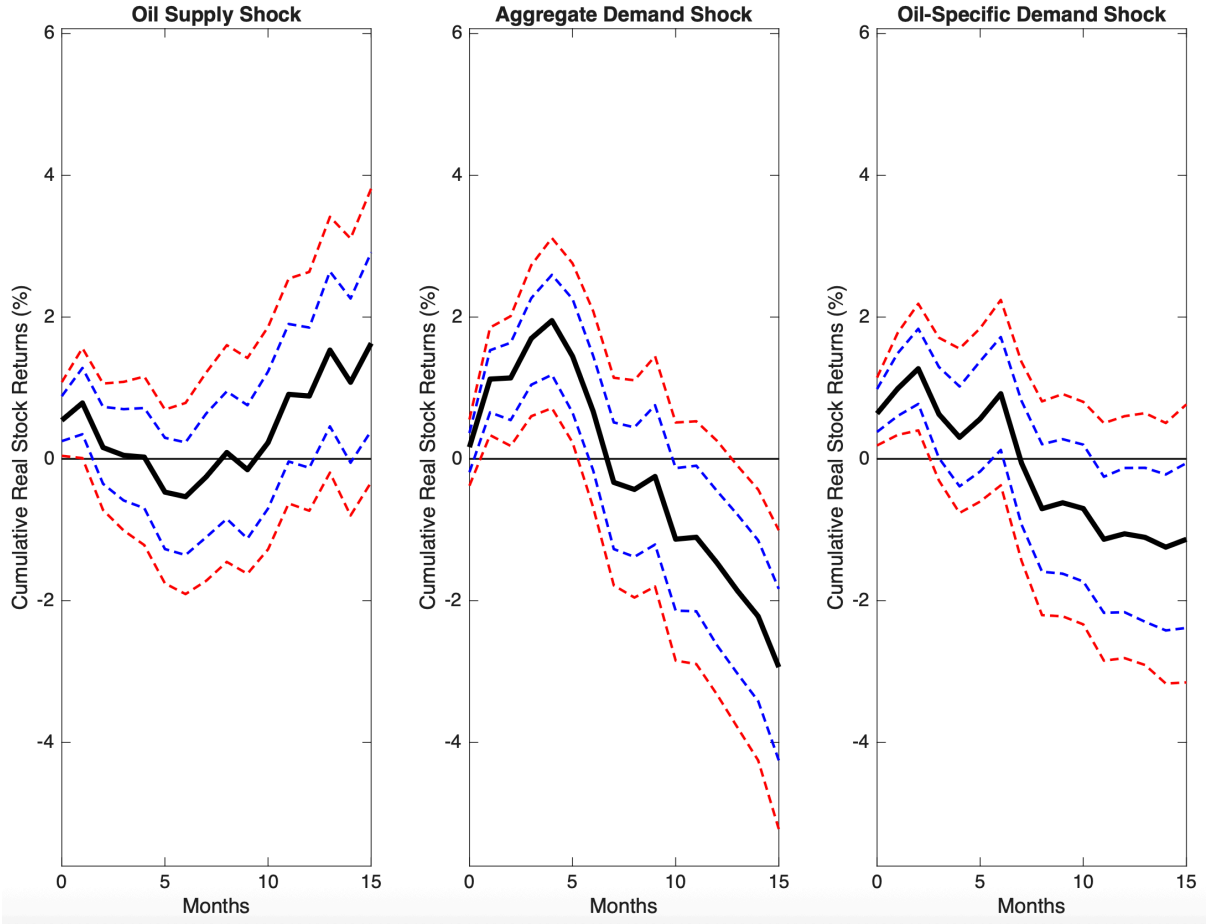
NOTES: Estimates based on the unrestricted structural VAR model.

TABLE 2
PERCENT CONTRIBUTION OF DEMAND AND SUPPLY SHOCKS IN THE CRUDE OIL MARKET TO THE OVERALL VARIABILITY OF SP500 REAL STOCK RETURNS

Horizon	Oil Supply Shock	Aggregate Demand Shock	Oil-specific Demand Shock	Other Shocks
1	2.09	5.60	3.13	89.18
2	4.27	5.41	3.48	86.84
3	4.16	6.89	5.58	83.37
12	7.38	16.29	11.41	64.92
∞	11.58	25.36	15.31	47.75

NOTES: Estimates based on the variance decomposition of the unrestricted structural VAR model using monthly data from 2006.12 to 2025.3.

FIGURE 10
CUMULATIVE RESPONSES OF SP500 REAL STOCK RETURNS TO ONE STANDARD DEVIATION STRUCTURAL SHOCKS: POINT ESTIMATES WITH ONE AND TWO STANDARD ERROR BANDS



NOTES: Estimates based on the unrestricted structural VAR model using monthly data from 2006.12 to 2025.3. Confidence bands were constructed using bootstrap-after-bootstrap (see Kilian 1998). Oil supply shock has been normalized to represent a negative one standard deviation shock.

There are several differences between the results in Figure 10 and those in Kilian and Park (2009), potentially reflecting different market conditions post-2006. For example, an aggregate demand shock causes a more immediate and substantial increase in U.S. real stock returns, followed by a sharper decline. Moreover, an oil-specific demand shock now leads to a gradual decline in U.S. real stock return, whereas pre-2009 reported persistently negative returns. Finally, an oil supply shock post-2006 leads to an initial fall in U.S. real stock return followed by a subsequent increase, while pre-2009 reported minor effects on these returns.

In the short run, the three structural shocks together explain about 11% of the variation in U.S. real stock returns, with aggregate demand shocks contributing the largest share (around 6% on impact). This was not the case pre-2006 when oil-specific demand shocks contributed the most, suggesting that the U.S. stock market has since become less sensitive to global oil market fluctuations in the short run. This is consistent with the gradual decline in Figure 10 as opposed to persistent negative returns pre-2006 following an oil-specific demand shock. In the long run, the three shocks together account for as much as 42% of the variation in U.S. real stock returns. The U.S. stock market continues to be demand driven, but much of this comes from aggregate demand shocks (25%), while oil supply and oil-specific demand shocks account for roughly 11% and 15% respectively.

5. Conclusion

This paper attempts to replicate the results in Kilian and Park (2009). Some of these results have been shown to vary depending on the model choice, the measures used, and the sample period. It should be noted, however, that different data sources, data availability, and data revisions may have impacted this paper's ability to replicate their findings.

The main finding in Kilian and Park (2009) – that structural shocks have different effects on the variability of U.S. stock returns – is robust under all conditions listed above. Oil supply and demand shocks do account for a fifth of the variation in stock returns within their sample period, and has further been shown to contribute as much as 42% in the extended sample. However, the effect of an aggregate demand shock on stock returns seem to vary depending on which model (structural VAR vs. Local Projection) was employed, which two stock return measures (S&P 500 vs. S&P 500 TR) were used, and which sample period (1973.12-2006.12 vs. 2006.12-2025.3) was chosen. Though, the response of oil prices to structural shocks were shown to be reasonably robust under these conditions.

References

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